

ELI ZAMAR BASHAN

DFIR Lead | AI Engineer

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PROFESSIONAL PROFILE

I am a cybersecurity professional with specialty in digital forensics and incident response with over eight years of experience handling high-impact security incidents, criminal investigations, and forensic matters with real-world consequences. I have served as a Security Operations Analyst with the FICSOC team supporting Ghana's banking sector, and I currently lead the forensic lab at Virtual InfoSec Africa, an MSSP serving clients across all 13 of Ghana's critical sectors. My experience includes preparing over 50 forensic and incident response reports, including legal reports used in court and police proceedings, serving as an expert witness, and building, training, and mentoring teams of analysts and responders. I have also designed and built production platforms across threat intelligence, forensic automation, OSINT investigation, governance and compliance, and AI-assisted health and safety management.

PLATFORMS BUILT

Production systems I designed and built independently, outside of employer-assigned work.

Heimdall *National Threat Intelligence Platform*

Ghana-focused threat intelligence platform aggregating IOCs, CVEs, threat actor TTPs, sector exposure data, and breach intelligence for national-level defenders. Includes an AI threat brief generator, semantic search over 8,500 indexed incidents, and a React-based analyst assistant with 8 specialized tools. Built for the West African threat landscape rather than a global feed calibrated for US and European environments.

Stack: Next.js, Python/FastAPI, PostgreSQL, Celery, Redis, Ollama, Claude API

Alfacotex *Automated DFIR Platform*

Consumer-facing DFIR platform, including a Tauri-based agent for local forensic collection on Windows and macOS, a YARA detection engine loading 1,500+ rules, an AI examination engine that builds forensic narratives and attack timelines from scan data, a visual attack map, a privacy scanning module, and a scheduled scan system. Evidence is structured into reports with an evidence meter and confidence layer designed to avoid false certainty.

Stack: Next.js, FastAPI, Rust (Tauri), PostgreSQL, YARA, Claude API, Docker

Kutarisa *OSINT Investigation Platform*

Link analysis and OSINT investigation platform built on React Flow with D3 force layout and ELK.js for graph arrangement, a recon API layer for entity enrichment, and a case management system structured around the investigation lifecycle from recon to case to report.

Stack: Next.js, React Flow, D3, FastAPI, PostgreSQL

VIA Assurance *GRC Platform*

ISO 27001, ISO 9001, and NIST CSF 2.0 governance, risk, and compliance platform built for VIA. Covers 17 compliance registers with a shared data view supporting table, card, and board layouts, bulk operations, export, grouping, and saved views. Includes audited write services with self-committing audit trails and role-based access control.

Stack: Next.js, FastAPI, PostgreSQL, Docker

SafelQOS *AI-Native Health and Safety SaaS*

Multi-tenant occupational health and safety platform for a UK-based client, live in production. Implements ISO 45001-aligned incident management with AI-assisted systemic investigation, hierarchy of controls, and CAPA workflows. Includes an AI privacy gateway that classifies and filters data before any model egress, preserving organizational data sovereignty.

Stack: Next.js, FastAPI, PostgreSQL, Ollama, Claude API, Docker

EXPERT WITNESS AND LEGAL CASE EXPERIENCE

Due to NDAs and the nature of the work, client and case details cannot be disclosed. The following describes the scope of the casework.

I have prepared many forensic and incident response reports, five of which were legal reports submitted in court or police proceedings, and I have served as an expert witness in one case, supporting a defensible interpretation of digital evidence and forensic methodology. The casework spans:

- Financial fraud and cybercrime, including business email compromise, online banking fraud, and cryptocurrency theft
- Ransomware attacks against corporate and institutional targets
- Insider threat investigations in banking and government environments
- Cyberbullying and online harassment cases
- Romance fraud and digital extortion
- Data breach investigations with regulatory reporting requirements
- DDOS and data destruction incidents

Across this casework I have managed chain of custody, maintained legal compliance, prepared court-ready evidence packages, and provided forensic opinion. The clients have included financial institutions, government agencies, educational institutions, and mining companies. The work spans mobile, computer, memory, network, cloud, and email forensics.

TECHNICAL SKILLS

Digital Forensics: Belkasoft, Cellebrite, Magnet Axion, Detego Forensic Suite, Autopsy, FTK, X-Ways, MD Next, Volatility 3, KAPE, Eric Zimmerman Tools, Elcomsoft, Passware, Scalpel, Foremost, DDRescue

Incident Response and Threat Intelligence: CyberTriage, Velociraptor, Arctic Security, InQuest, LogRhythm, Exabeam, AlienVault, EventTracker, Kaspersky EDR, CrowdStrike, Cortex XDR, EDR/XDR platforms, SIEM investigation

Malware Analysis: IDA Pro, Ghidra, YARA rule authoring and engine integration, PsExec

OSINT: Maltego, OSINT Framework, passive DNS, BGP/ASN analysis, breach intelligence feeds

AI Engineering: Claude API, Ollama local LLM deployment, RAG pipeline design, ReAct agent architecture, prompt engineering, AI privacy gateway design, semantic search with vector embeddings

Software Development: Python, FastAPI, Next.js/React, PostgreSQL, Docker, Redis, Celery, Rust (Tauri), Bash, PowerShell

Penetration Testing: Metasploit, Burp Suite, Nmap

Infrastructure: Linux, Windows Server, VPS deployment, Nginx, cloud forensics on AWS and Azure

PROFESSIONAL EXPERIENCE

DFIR Lead | Virtual InfoSec Africa, Accra, Ghana

February 2025 to Present

- Lead the forensic lab within an MSSP serving many clients across all 13 of Ghana's critical sectors, including banking, finance, government, telecommunications, energy, and healthcare
- Coordinate with the FICSOC team that provides security oversight for Ghana's 24 member banks, handling cases where security incidents require forensic follow up
- Direct forensic investigations and incident response on criminal and civil cases with legal evidentiary requirements
- Oversee deployment of enterprise forensic tooling including Detego, Elcomsoft, Passware, Belkasoft, Magnet Axiom, Cellebrite, FTK, and Volatility
- Lead OSINT investigations and threat intelligence operations
- Manage evidence handling, chain of custody, and legal compliance on high-stakes casework
- Train and mentor junior DFIR analysts and establish team SOPs, workflows, and playbooks

Digital Forensic Analyst, Level 1 | Virtual InfoSec Africa, Accra, Ghana

January 2024 to January 2025

- Forensic analysis across mobile, memory, network, and computer environments
- OSINT, threat intelligence, and incident response support

Digital Forensics Examiner / Consultant | Self-Employed, Ghana

August 2022 to December 2023

- Independent DFIR consulting for banking, government, education, and mining clients
- Vulnerability assessments, system hardening, and forensic data recovery

Volunteer Contributor | Atomic Red Team Community, Remote, US

January 2023 to December 2023

- Contributed bug reports and MITRE ATT&CK-aligned script improvements to the global open-source red teaming project

Cybersecurity Administrator | Virtually Testing Foundation, Remote, US

January 2023 to March 2023

- Vulnerability assessments, OSINT investigations, and security training delivery

Laboratory / XRF Technician, IT Support | Sahara Natural Resources, Ghana

March 2022 to August 2023

- Managed IT infrastructure and cybersecurity for laboratory operations

Information Systems Engineer | Sefwi Wiawso Nursing Training School, Ghana

September 2019 to September 2020

- Installed and managed a 150+ workstation lab on Windows Server with network monitoring
- Built a student database for 700+ students

KEY ACHIEVEMENTS

- Lead the forensic lab at an MSSP with the broadest DFIR mandate of its kind in Ghana, covering many clients across all 13 national critical sectors, alongside a SOC providing oversight for 24 member banks
- Prepared over 50 forensic and incident response reports, including 5 legal reports submitted in court and police proceedings
- Served as an expert witness in one case, supporting the courtroom interpretation of digital evidence and forensic methodology

- Supported a hospitality sector engagement that prevented more than USD 240,000 in potential loss through DFIR-led recovery and response
- Trained over 100 professionals and students in digital forensics and incident response, and mentored many junior analysts
- Designed and shipped five production platforms independently, across DFIR, threat intelligence, OSINT, GRC, and AI-assisted health and safety
- Trained by UNODC and Interpol in ransomware investigation and cross-border cybercrime
- Delivered DFIR training at KNUST, IDL, and Palmers University
- Designed the Integrated IoT Forensic Investigation Framework (I2FIF)
- Contributed to Atomic Red Team, an open-source project used by security teams globally
- Published Writing Three articles in the Business and Financial Times (Ghana):
 - "The Cryptocurrency Conundrum"
 - "Unravelling Tomorrow's Cyber Saga: Anticipating Current and Future Threats"
 - "Exploring the Tech Revolution"

EDUCATION

- MSc, Cybersecurity and Digital Forensics, Kwame Nkrumah University of Science and Technology (KNUST)
- BSc (Hons), Computer Science, Kwame Nkrumah University of Science and Technology (KNUST)
- Diploma, Theological and Ministerial Studies, Cave Adullam Bible Seminary

CERTIFICATIONS

- Cellebrite Mobile Forensic Examiner (WeCheck)
- Detego Certified Forensic Examiner
- EC-Council Certified Incident Handler
- EC-Council Certified Security Operations Center Analyst
- EC-Council Certified Security Analyst
- LogRhythm Certified Security Analyst
- LogRhythm Certified Platform Administrator
- Cyberwarfare Labs Certified Cyber Security Analyst
- TCM Security Practical Windows Forensics
- Certified CyberArk Trustee
- Fortinet Certified Associate Cybersecurity
- Fortinet Network Security Expert, Level 1
- Fortinet Network Security Expert, Level 2
- ISO/IEC 27001 Information Security Associate
- ISO/IEC 20000 IT Service Management Associate
- Certiprof Certified Cybersecurity Awareness Professional

References and published work available on request. Expert witness methodology and casework can be discussed under NDA.